

Áp dụng (Apply It 2.2) - Phân tích Bán hàng (Super Scooters)

AutoSave Off										
	A	B	C	D	E	F	G	H	I	
1	OrderNumber	Model	SoldDate	SalesVolume	Color	Location	State	DaysInInventory	UnitSalePrice	G
2	13684	Celeritas	12/31/2025	20	Yellow	Dallas	TX	83	\$ 342.00	
3	13685	Lazer	12/31/2025	20	Blue	Seattle	WA	52	\$ 414.00	
4	13682	Captain	12/30/2025	20	Green	Seattle	WA	29	\$ 679.00	
5	13683	Lazer	12/30/2025	38	Green	Miami	FL	22	\$ 376.00	
3645	10957	Lazer	12/22/2023	18	Yellow	Phoenix	AZ	17	\$ 330	
3646	10960	Lazer	12/31/2023	35	Red	Phoenix	AZ	24	\$ 357	
3647										
Sheet1										

Hình: Sử dụng VLOOKUP, SUM, SUMIF để tổng hợp dữ liệu giao dịch.

2.3 Tổ Chức Các Tập Dữ Liệu: Pivot Tables

- **PivotTable** là một công cụ tương tác mạnh mẽ dùng để tóm tắt và phân tích lượng dữ liệu khổng lồ.
- 4 Vùng làm việc chính:
 - **Rows (Hàng):** Nhóm dữ liệu theo danh mục theo chiều dọc.
 - **Columns (Cột):** Nhóm dữ liệu theo danh mục theo chiều ngang.
 - **Values (Giá trị):** Thực hiện tính toán (SUM, COUNT, AVERAGE,...) trên các trường số liệu.
 - **Filters (Bộ lọc):** Lọc toàn bộ PivotTable theo các điều kiện.

Tạo một Microsoft Excel PivotTable

ILLUSTRATION 2.17 University Assets Data Excel Spreadsheet

AssetTagID	AssetDescription	CategoryID	Category	Amount	AcquisitionDate
U000009	Pump	15	Laboratory Equipment	\$ 7,826.00	9/4/2015
U000010	Bonder	15	Laboratory Equipment	\$ 28,628.01	9/13/2015
U000015	Irrigation System	1	Agricultural and Farm Machinery	\$ 17,068.00	10/1/2015
U000016	Drill	1	Agricultural and Farm Machinery	\$ 28,344.00	10/29/2015
U000018	Video Camera	6	Communication, Audio-Visual, Alarm, & Signal Equipment	\$ 33,573.00	9/9/2015
U000020	Ultrasound System	15	Laboratory Equipment	\$ 13,720.00	9/24 /2015
U000022	Pump	15	Laboratory Equipment	\$ 13,552.65	9/9/2015

Hình: Chọn Insert > PivotTable.

ILLUSTRATION 2.18 Excel Create PivotTable Dialog Box

PivotTable from table or range

Select a table or range

5 Table/Range: 'University Inventory Data'!\$A\$1:\$F\$13127

Choose where you want the PivotTable report to be placed

☒ New Worksheet

☐ Existing Worksheet

Location:

Choose whether you want to analyze multiple tables

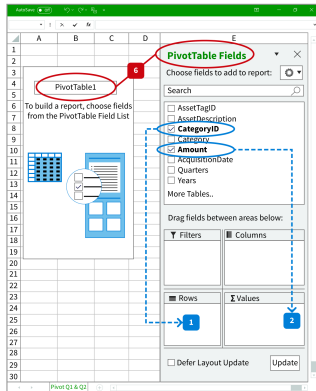
☐ Add this data to the Data Model

OK Cancel

Hình: Chọn vùng dữ liệu nguồn.

Vùng làm việc PivotTable (PivotTable Fields)

ILLUSTRATION 2.19 Blank PivotTable Canvas



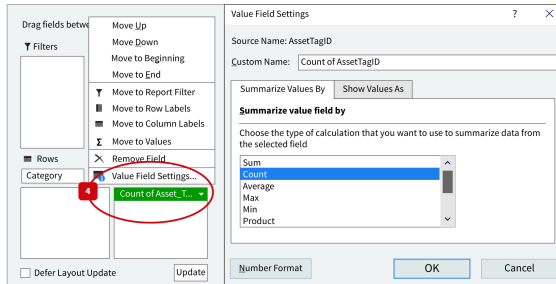
Hình: Danh sách các trường (Fields) và 4 vùng (Areas) kéo thả.

Tìm Tổng Số dư theo Danh mục (Summarize Data)

ILLUSTRATION 2.20 Excel PivotTable for Question 1: What is the Total Balance by Category?

Question 1	
Row Labels	Sum of Amount
Agricultural and Farm Machinery	\$ 3,838,203.58
Art Equipment	\$ 129,924.35
Athletic and Recreational Equipment	\$ 1,047,971.77
Boats, Boat Motors, and Marine Equipment	\$ 530,673.88
Cleaning and Maintenance Equipment	\$ 2,458,319.00
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 4,242,923.45
Computer Equipment	\$ 43,294,238.92
Drafting and Surveying Equipment	\$ 18,033.00
Electrical Appliances and Equipment	\$ 51,825.00
Electronic Equipment	\$ 278,233.02
Food Preparation and Serving Equipment	\$ 1,210,703.93
Furniture and Furnishings	\$ 3,060,781.37
Hand and Power Tools (Portable)	\$ 90,332.00
Industrial, Shop, and Construction Equipment	\$ 2,287,996.49
Laboratory Equipment	\$ 147,343,867.13
Material Handling Equipment	\$ 650,212.23
Medical Equipment	\$ 2,449,781.57
Miscellaneous Equipment	\$ 47,239.00
Motor Vehicles and Transportation Equipment	\$ 1,270,121.50
Musical Instruments	\$ 88,248.00
Office and Business Machines and Non-Consumable Supplies	\$ 1,587,495.45
Parking Control Equipment	\$ 230,506.72
Photographic Equipment	\$ 1,865,573.30

ILLUSTRATION 2.21 Excel PivotTable Value Field Settings Dialog Box



Hình: PivotTable hiển thị Tổng chi phí theo

Thay đổi Phép tính (Value Field Settings)

ILLUSTRATION 2.22 Excel PivotTable for Question 2: What is the Total Count of Assets in Each Category?

Row Labels	Question 1	Question 2
	Sum of Amount	Count of AssetTagID
Agricultural and Farm Machinery	\$ 3,838,203.58	171
Art Equipment	\$ 129,924.35	7
Athletic and Recreational Equipment	\$ 1,047,971.77	80
Boats, Boat Motors, and Marine Equipment	\$ 530,673.88	37
Cleaning and Maintenance Equipment	\$ 2,458,319.00	182
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 4,242,923.45	271
Computer Equipment	\$ 43,294,238.92	2543
Drafting and Surveying Equipment	\$ 18,033.00	2
Electrical Appliances and Equipment	\$ 51,825.00	5
Electronic Equipment	\$ 278,233.02	29
Food Preparation and Serving Equipment	\$ 1,210,703.93	112
Furniture and Furnishings	\$ 3,060,781.37	185
Hand and Power Tools (Portable)	\$ 90,332.00	5
Industrial, Shop, and Construction Equipment	\$ 2,287,996.49	154
Laboratory Equipment	\$ 147,343,867.13	8057
Material Handling Equipment	\$ 650,212.23	47
Medical Equipment	\$ 2,449,781.57	113
Miscellaneous Equipment	\$ 47,239.00	4
Motor Vehicles and Transportation Equipment	\$ 1,270,121.50	75
Musical Instruments	\$ 88,248.00	7
Office and Business Machines and Non-Consumable Supplies	\$ 1,587,495.45	83
Parking Control Equipment	\$ 230,506.72	12
Photographic Equipment	\$ 1,865,573.30	143
Plumbing, Heating, Air Conditioning, and Ventilating Equipment	\$ 146,953.15	17
Police, Fire Fighting, and Safety Equipment	\$ 102,458.55	12
Printing and Bookbinding Equipment	\$ 279,665.97	19
Refrigeration Equipment	\$ 6,467,000.41	754
Grand Total	\$ 225,069,282.74	13126

PivotTable Fields

Choose fields to add to report

Search

☒ AssetTagID
☐ AssetDescription
☐ CategoryID
☒ Category
☒ Amount
☐ AcquisitionDate
☐ Quarters
☐ Years

More Tables..

Drag fields between areas below:

Filters

Columns

Rows

Values

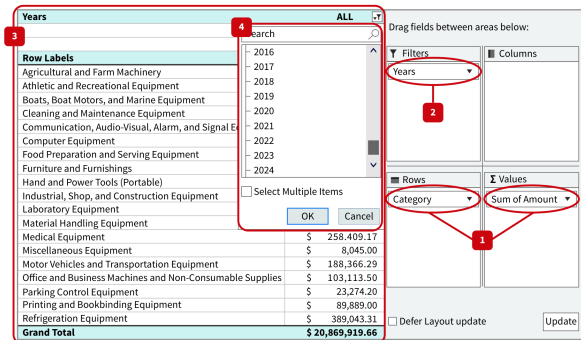
Sum of Amount
Count of AssetTagID

☐ Defer Layout Update

Hình: Thay đổi hàm từ Sum thành Count để đếm số lượng tài sản.

Lọc Pivot Tables (Hộp Filter)

ILLUSTRATION 2.23 How to Filter in an Excel PivotTable



Hình: Kéo trường Asset Sub-Category vào vùng Filters để tạo bộ lọc tổng.

Sử dụng Slicers để Lọc Trực quan

ILLUSTRATION 2.32 Adding Slicers to PivotTables

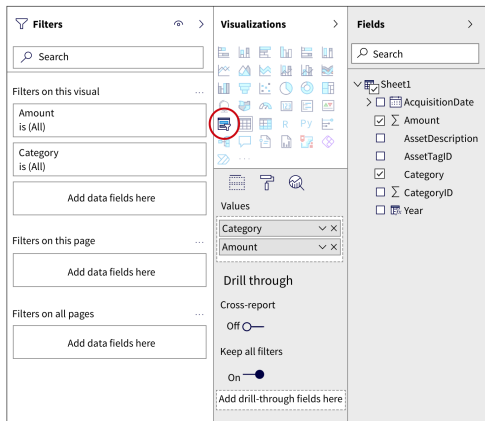
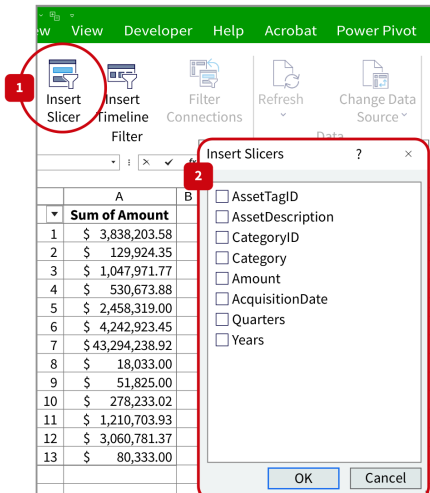


ILLUSTRATION 2.34 Pivot Table Slicers Using PowerBI

Hình: Slicer tạo các nút bấm trực quan để lọc dữ liệu

Áp dụng (Apply It 2.3) - Phân tích Bán hàng bằng PivotTables

Year	2025 
Row Labels 	Sum of Gross Sales
Boston	\$ 1,433,578.00
Charlotte	\$ 1,653,506.00
Chicago	\$ 1,624,719.00
Dallas	\$ 1,134,785.00
Miami	\$ 1,606,576.00
Phoenix	\$ 1,376,503.00
Salt Lake City	\$ 1,593,184.00
Seattle	\$ 1,742,311.00
Grand Total	\$ 12,165,162.00

Hình: Tạo báo cáo doanh số theo tháng và sản phẩm.

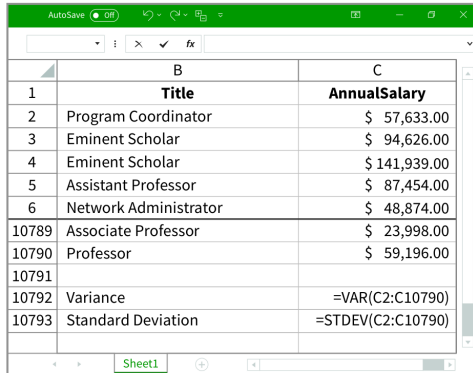
2.4 Thước Đo Mô Tả Giúp Hiểu Dữ Liệu

- Thống kê mô tả được sử dụng trong giai đoạn **Khám phá Dữ liệu** (Giai đoạn 2).
- 3 nhóm thước đo chính:
 - **Vị trí trung tâm (Location/Central Tendency):** Mean (Trung bình), Median (Trung vị), Mode (Yếu vị).
 - **Độ phân tán (Dispersion):** Range (Khoảng), Variance (Phương sai), Standard Deviation (Độ lệch chuẩn).
 - **Hình dạng (Shape):** Skewness (Độ lệch), Kurtosis (Độ nhọn).

Tính toán Các Thước đo Vị trí (Mean, Median, Mode)

ILLUSTRATION 2.38 University
Payroll Data

University Payroll Data



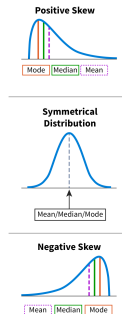
The screenshot shows an Excel window with the title 'University Payroll Data'. The spreadsheet has two columns: 'Title' (Column B) and 'AnnualSalary' (Column C). The data rows are numbered 1 through 6, then 10789, 10790, 10791, 10792, and 10793. The last three rows contain summary statistics: Variance and Standard Deviation, calculated using Excel formulas.

	B	C
	Title	AnnualSalary
1		
2	Program Coordinator	\$ 57,633.00
3	Eminent Scholar	\$ 94,626.00
4	Eminent Scholar	\$ 141,939.00
5	Assistant Professor	\$ 87,454.00
6	Network Administrator	\$ 48,874.00
10789	Associate Professor	\$ 23,998.00
10790	Professor	\$ 59,196.00
10791		
10792	Variance	=VAR(C2:C10790)
10793	Standard Deviation	=STDEV(C2:C10790)

Hình: Sử dụng hàm AVERAGE, MEDIAN, MODE.SNGL trong Excel.

Các thước đo Hình dạng (Skewness)

ILLUSTRATION 2.41 Skewness
Examples



Hình: Phân phối lệch trái (Left-Skewed) và lệch phải (Right-Skewed).

Biểu đồ Tần suất (Histograms)

ILLUSTRATION 2.43 Excel Data Analysis Tool

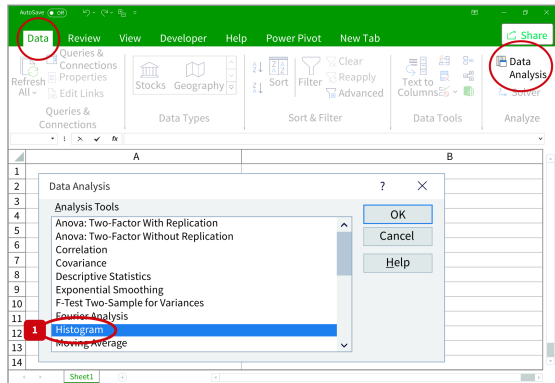
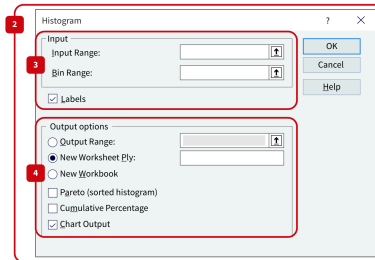


ILLUSTRATION 2.44 Histogram Dialog Box



Hình: Trực quan hóa bằng Histogram.

Hình: Xây dựng bảng phân phối tần số (Bins).

Các Công cụ Thống kê Mô tả Excel (Data Analysis Toolpak)

ILLUSTRATION 2.45 Payroll Data, Bins, and Histogram Dialog Box

Payroll Data, Bins, and Histogram Dialog Box

	B	C	D	E
1	Title	Annual Salary		Bins
2	Program Coordinator	\$ 57,633.00		8,000
3	Eminent Scholar	\$ 94,626.00		20,000
4	Eminent Scholar	\$ 141,839.00		40,000
5	Assistant Professor	\$ 87,454.00		60,000
6	Network Administrator	\$ 45,874.00		80,000
7	Student Program Coordinator	\$ 42,630.00		100,000
8	Professor	\$ 113,736.00		120,000
9	Professor	\$ 2,170.00		140,000
10	Professor	\$ 115,228.00		160,000
11	Research Assistant	\$ 6,700.00		180,000
12	Research Assistant	\$ 6,700.00		200,000
13	Research Assistant	\$ 20,100.00		220,000
14	Professor	\$ 119,900.00		
10789	Associate Professor	\$ 23,998.00		
10790	Professor	\$ 99,196.00		

Histogram

Input

Input Range:

Bin Range:

☒ Labels

Output options

☐ Output Range:

☒ New Worksheet Ply:

☐ New Workbook

☐ Pareto (sorted histogram)

☐ Cumulative Percentage

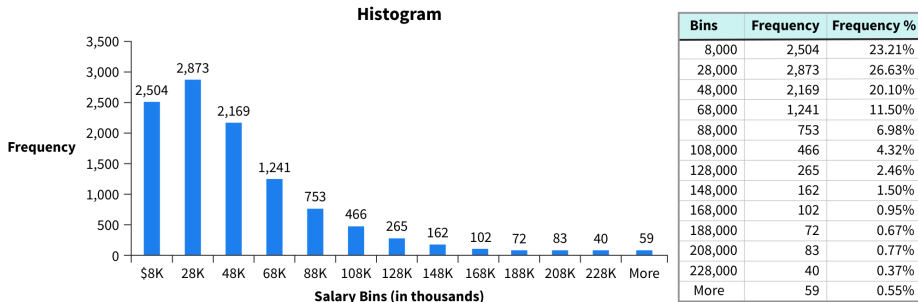
☒ Chart Output

OK Cancel Help

Hình: Tự động xuất toàn bộ các chỉ số thống kê cơ bản trong 1 cú nhấp chuột.

Phân tích Tương quan (Correlation Analysis)

ILLUSTRATION 2.46 University Payroll Data Histogram and Frequency Distribution



Hình: Hệ số tương quan (r) đo lường mức độ liên hệ giữa 2 biến liên tục (từ -1 đến 1).